ECON 470 Homework 3

Submission 3, Spring 2026

Jane Miller

```
data.2010 <- read.csv('../data/data-2010.csv')
data.2011 <- read.csv('../data/data-2011.csv')
data.2012 <- read.csv('../data/data-2012.csv')
data.2013 <- read.csv('../data/data-2013.csv')
data.2014 <- read.csv('../data/data-2014.csv')
data.2015 <- read.csv('../data/data-2015.csv')
```

```
fix_type <- function(df) {
  df$org_parent <- as.character(df$org_parent)
  df
}

data.2011 <- fix_type(data.2011)
```

```
data.full <- bind_rows(data.2010, data.2011, data.2012, data.2013, data.2014, data.2015)  %>%
  mutate(market_share = avg_enrollment / avg_enrolled)
```

Problem 1

Table 1: Summary Statistics by Year

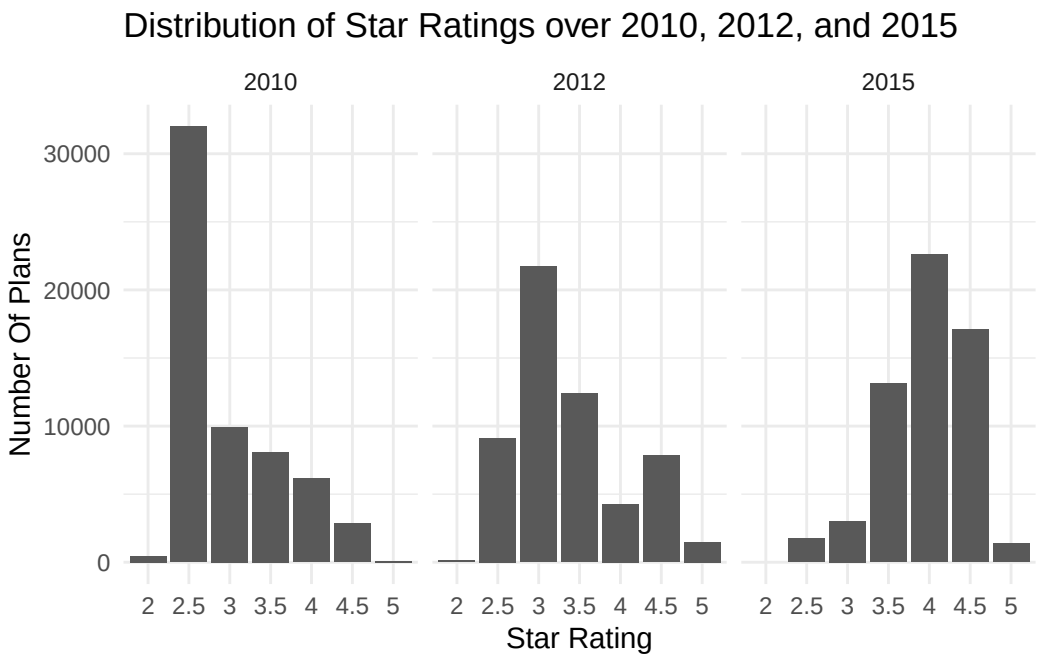
Year	Avg Star Rating	Avg Enrollments	Avg Market Share	Total Plans
2,010	2.97	256.35	0.0658	108,222
2,011	3.26	343.67	0.0857	66,544
2,012	3.36	377.68	0.0858	65,891
2,013	3.55	395.93	0.0817	66,067
2,014	3.81	436.87	0.0802	60,497
2,015	3.96	470.17	0.0789	63,626

Problem 2

Table 2: Summary Statistics for Plans Without a Star Rating

Year	Avg Enrollments	Avg Market Share	Total Plans
2,010	107.34	0.0673	48,643
2,011	167.81	0.0904	12,458
2,012	172.89	0.0569	8,908
2,013	212.71	0.0602	3,585
2,014	255.04	0.0562	3,664
2,015	314.74	0.0407	4,629

Problem 3



We note that there is a steep decrease in the number of high-rated plans in 2010, whereas in 2012, the difference is less stark, with improvement in 3-star-rated plans, and in 2015, we notice an improvement, with the existence of a larger number of high-rated plans.

Problem 4

Table 3: OLS Estimates of Star Rating Effect on Market Share by Year

	2010	**2011**	**2012**	**2013**	**2014**	**2015**
Reference: <= 2.5 Stars						
(Intercept)	0.0634	0.0703	0.0791	0.0490	0.0640	0.0606
Star Rating Indicators						
star_3	0.0126	0.0350	0.0307	0.0624	0.0269	0.0272
star_35	0.0005	0.0192	0.0129	0.0387	0.0275	0.0389
star_4	-0.0036	0.0026	-0.0247	0.0172	0.0076	0.0115
star_45	-0.0076	-0.0094	-0.0134	0.0225	0.0072	0.0234

Problem 5

Here, we create a raw rating as a simple average or an aggregate of all important factors.

Table 4: Number of Plans Rounded Up to Each Star Rating (2010)

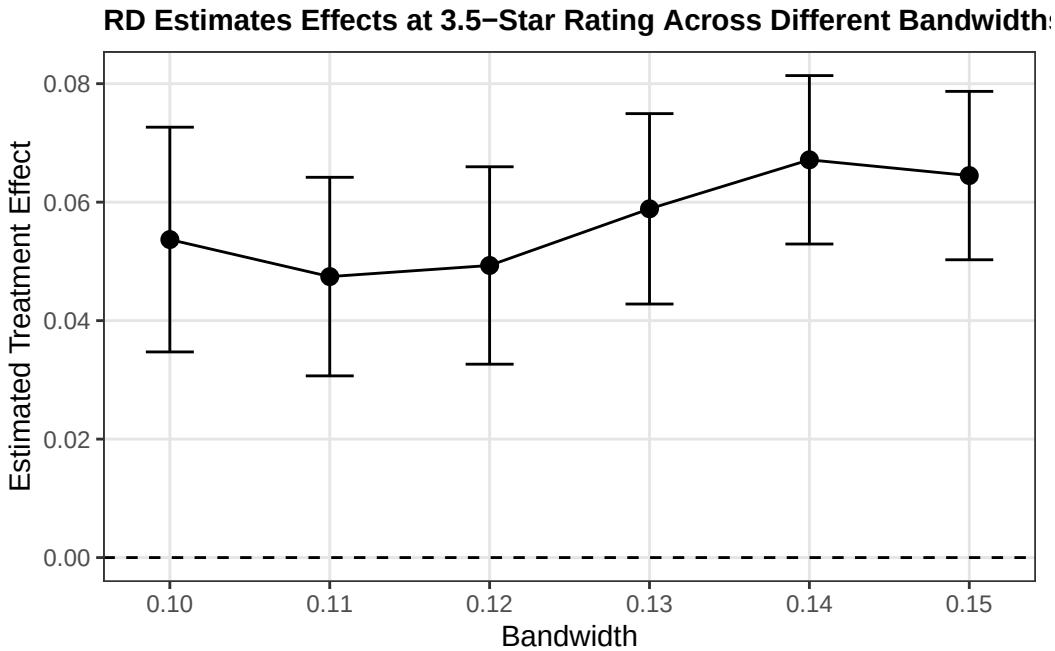
Star Rating	Number of Plans
Rounded Up to 3	1,672
Rounded Up to 3.5	1,762
Rounded Up to 4	1,594
Rounded Up to 4.5	193
Rounded Up to 5	29

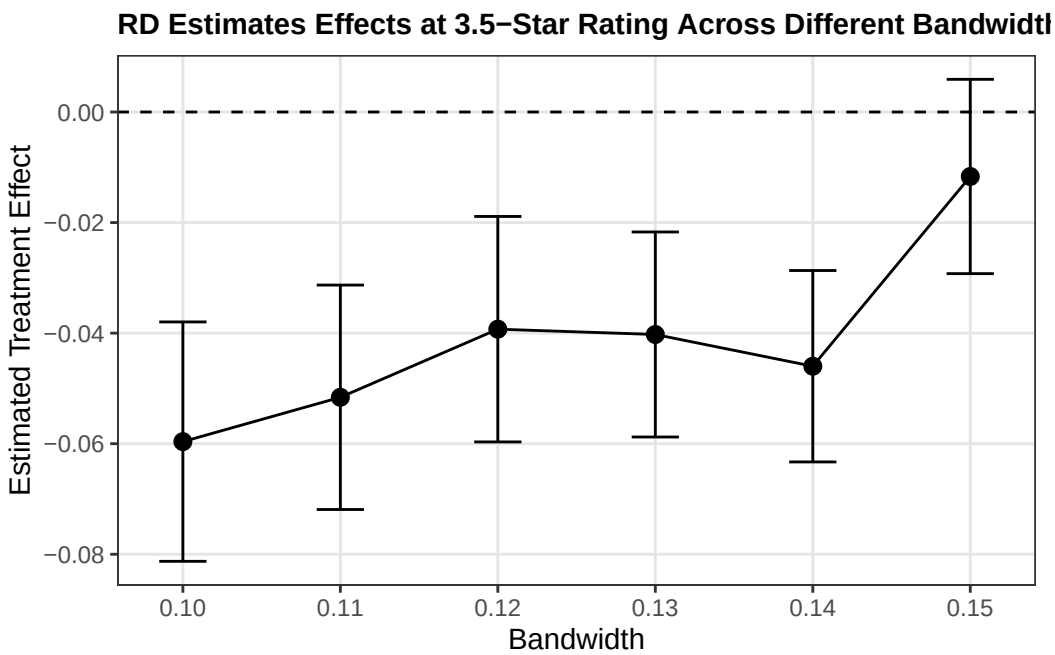
Problem 6

Table 5: RD Estimates at 3-Star and 3.5-Star Thresholds

Statistic	3 Stars	3.5 Stars
Coefficients		
Intercept	0.016	0.102
Rounded	0.055	-0.040
Running Score	-0.740	0.417
Interaction	0.958	-0.467
Model Fit		
Num.Obs.	5309.000	1882.000
R2	0.059	0.015

Problem 7

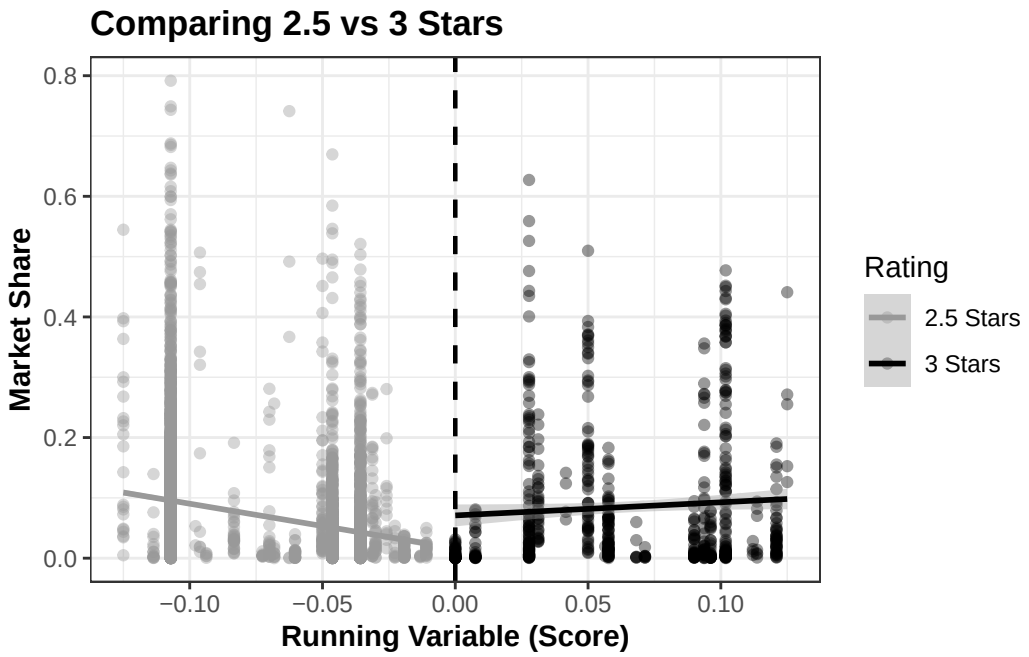


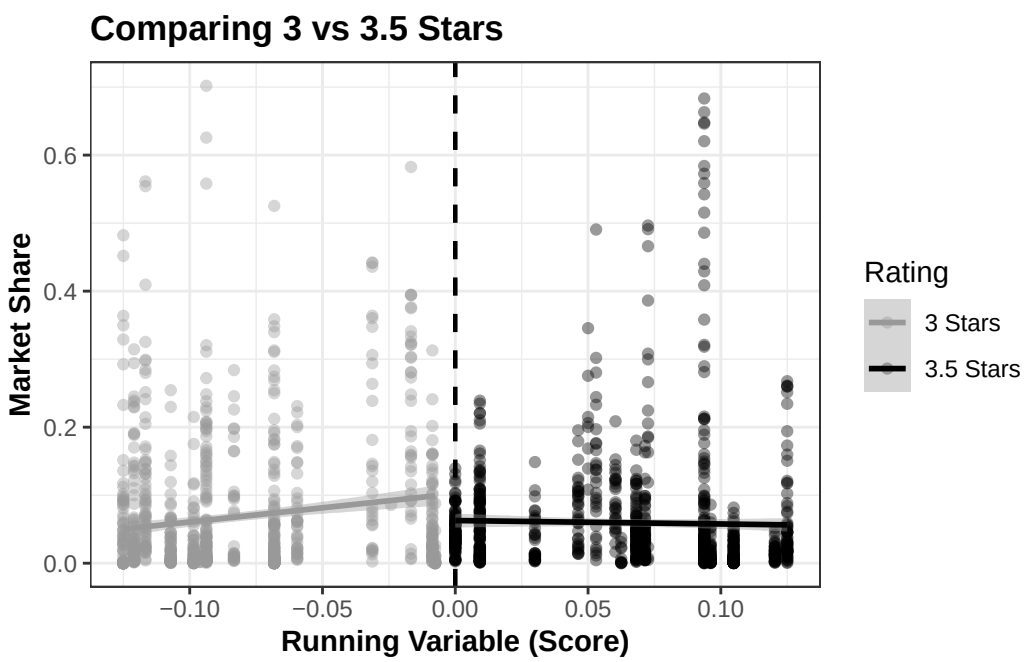


We note that findings are indeed sensitive to bandwidth choice. For instance, the treatment effect in comparing 3 versus 3.5 star plans appears to be more vivid when compared at the threshold of 0.1 as opposed to 0.15. Similarly, we see a difference in calculated effect when a

bandwidth of 0.12 is chosen as opposed to 0.14 while comparsing 2.5 star plans versus 3-star plans.

Problem 8





We do see an abrupt increment in market share concording to values below and above the threshold, particularly prominent near the threshold of comparison for 2.5 stars against 3 stars (i.e., the first plot), particularly the drop from -0.10 to -0.05 in the upper plot. In contrast to this, we see a marginal increase in the number of plans from -0.10 to -0.05 in the

lower plot. All in terms of the accumulated frequency of the dotted columns, showing market share in means of the number of plans.

Problem 9

Table 6: Plan Characteristics Near The 2.75 Star Rating Margin

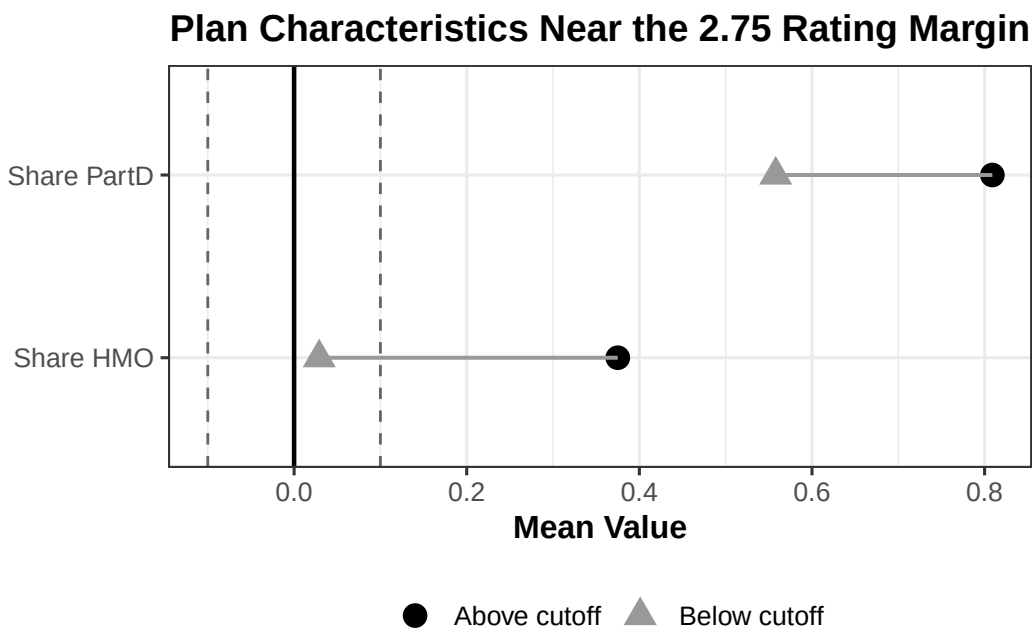
Group	Share HMO	Share Part D
Above Cutoff	0.375	0.809
Below Cutoff	0.029	0.558

Table 7: Plan Characteristics Near The 3.25 Star Rating Margin

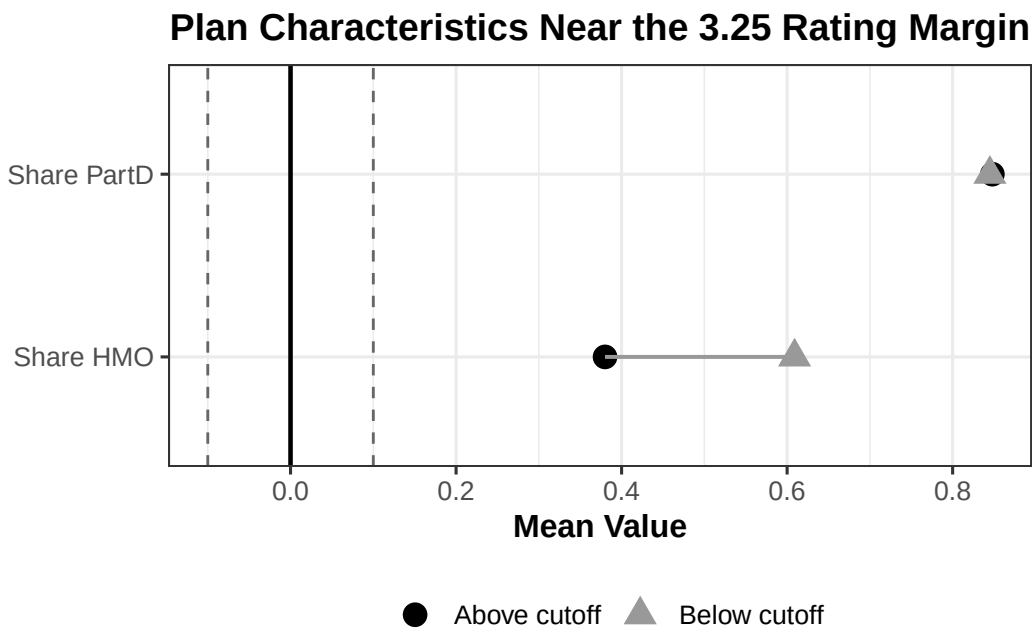
Group	Share HMO	Share Part D
Above Cutoff	0.380	0.848
Below Cutoff	0.609	0.845

```r
love_threshold_275 <- tibble(
  variable = rep(c("Share HMO", "Share PartD"), each = 2),
  group = rep(c("Above cutoff", "Below cutoff"), 2),
  value = c(0.375, 0.0291, 0.809, 0.558)
)

love_threshold_325 <- tibble(
  variable = rep(c("Share HMO", "Share PartD"), each = 2),
  group = rep(c("Above cutoff", "Below cutoff"), 2),
  value = c(0.380, 0.609, 0.848, 0.845)
)

love_plot <- function(data, title) {
  ggplot(data, aes(x = value, y = variable, color = group, shape = group)) +
    geom_point(size = 4) +
    geom_line(aes(group = variable), color = "grey60", linewidth = 0.7) +
    geom_vline(xintercept = 0, linewidth = 0.8) +
    geom_vline(xintercept = c(-0.1, 0.1), linetype = "dashed", color = "grey40") +
    scale_color_manual(values = c("Above cutoff" = "black", "Below cutoff" = "grey60")) +
    scale_shape_manual(values = c("Above cutoff" = 16, "Below cutoff" = 17)) +
    labs(
      title = title,
      x     = "Mean Value",
      y     = NULL,
      color = NULL,
      shape = NULL
    ) +
```

```
    theme_bw(base_size = 12) +
    theme(
      legend.position = "bottom",
      plot.title      = element_text(hjust = 0.5, face = "bold"),
      axis.title      = element_text(face = "bold")
    )
}

p1 <- love_plot(love_threshold_275, "Plan Characteristics Near the 2.75 Rating Margin")
p2 <- love_plot(love_threshold_325, "Plan Characteristics Near the 3.25 Rating Margin")
p1
```




```
p2
```



N signifies the number of plans. So, with a bandwidth of 0.125, we note that there is a substantially higher proportion of HMO-approved and Part D plans above the threshold for the nearest half-star comparison between 2.5 and 3-star ratings. Whereas, we see more HMO-approved plans below the threshold while comparing the nearest half-star for 3 against 3.5-star rated plans. But we see almost an equivalent proportion of Part D plans above and below the threshold. The proportions don't directly add to one row or column-wise since we're considering dummy fractions out of all possible plan categories that could be above or below the threshold.

Problem 10

The effect of increased star rating is directly proportional to a larger number of plans, particularly prominent from the upward and downward slopes, above and below the threshold, respectively, in the Problem 8 plot comparing 2.5-star plans against 3-star ones. The effect is marginal beyond 3 stars increase upto the half-star between 3 vs 3.5-star plans. This makes sense as plans try to achieve a better rating for more enrollments. However, these results are highly sensitive to the chosen bandwidth, as evident from Problems 6 & 7. Further, an effect is added from the fact that the 3-star ratings experience the largest number of round-ups as compared to plans above it, thereby making that region particularly volatile to analysis.

The increased plan count with the effect of having a higher rating also becomes prominent as we see increased Part D contained plans for above the threshold of 2.5 vs 3 star comparison, with a marginal incremental effect going from 3 vs 3.5 star. Similarly, we see more HMO-aligned plans above the threshold of 2.5 vs 3 but under the threshold of 3 vs 3.5, showing accumulation and tending effect of more plan count with HMO benefits as ratings increase, since HMO indicates lower out-of-pocket costs and premiums.